

ACTR-50. EFFECT OF CONCURRENT AND ADJUVANT TEMOZOLOMIDE ON SURVIVAL IN PATIENTS WITH NEWLY DIAGNOSED GRADE III GLIOMAS WITHOUT 1p/19q CO-DELETION: A RANDOMIZED, OPEN-LABEL, PHASE 2 STUDY (INTERIM RESULTS FROM THE KNOG-1101 STUDY)

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OBJECTIVE: We investigated the efficacy between radiotherapy alone versus concurrent chemoradiotherapy with temozolomide followed by adjuvant temozolomide in Korean adult patients with newly diagnosed non-co-deleted grade III gliomas. **METHODS:** This was a randomized, open-label, phase 2 study, and the first multicenter, prospective clinical trial for Korean patients with glioma. Eligible patients were aged 18 years or older and had newly diagnosed grade III gliomas with ECOG performance status of 0–2. No co-deletion of chromosome 1p/19q and further genetic biomarker was identified for available subjects. Patients were enrolled at 11 centers, and randomized 1:1 to receive radiotherapy alone (daily fractions of 2 Gy given 5 days per week for 6 weeks, for a total of 60 Gy) [Control group (n=44)], or to receive radiotherapy with daily temozolomide (75 mg/m²/day, 7 days/week from the first to the last day of radiotherapy) followed by six cycles of adjuvant temozolomide (150 to 200 mg/m²/day for 5 days during each 28-day cycle) [Treatment group (n=40)]. The primary endpoint was 2-year progression-free survival (PFS). **RESULTS:** At the time of the preliminary analysis, PFS at 2 years was 42.2% with treatment group and 37.2% with control group. On univariable analysis, extent of tumor resection, age, and IDH-1 mutant were significantly associated with PFS. On multivariable analysis, IDH-1 mutant was the only significant prognostic factor for PFS (HR0.27; 95% CI, 0.12–0.59; p=0.001). IDH-1 mutant was also the only independent prognostic factors for overall survival (HR0.10; 95% CI, 0.02–0.41; p=0.002). Adverse events over grade 3 were seen in 16 (40.0%) patients with Treatment group, but were mild and reversible. **CONCLUSIONS:** Radiotherapy with concurrent and adjuvant temozolomide in patients with newly diagnosed non-co-deleted anaplastic gliomas in Korean adults trended toward improved 2-year PFS. The survival benefit of this regimen would need further analysis for long-term follow-up.

ACTR-51. PHASE 2 STUDY TO EVALUATE THE SAFETY, PHARMACOKINETICS AND CLINICAL ACTIVITY OF PI3K/MTOR INHIBITOR GDC-0084 GIVEN TO GLIOBLASTOMA (GBM) PATIENTS WITH UNMETHYLATED O₆-METHYLGUANINE-METHYLTRANSFERASE PROMOTER STATUS

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BACKGROUND: GDC-0084 is a potent, oral, selective small molecule inhibitor of class I phosphoinositide 3-kinase and mammalian target of rapamycin (PI3K/mTOR) efficacious in GBM models driven by activation of the PI3K pathway. GDC-0084 crosses the blood-brain barrier (BBB) and achieves a brain / plasma ratio of approximately 1.0 in three animal species. GDC-0084 was given as once daily oral dosing in a phase 1 study (Wen et al, J Clin Oncol 34, 2016(15) suppl.2012; NCT01547546) in 47 patients with recurrent high-grade gliomas. The adverse events were generally consistent with the established Class I PI3K/mTOR inhibitor class-effects. GDC-0084 was rapidly absorbed and demonstrated linear- and dose-proportional increases in exposure. The MTD was determined to be 45 mg once daily, with 7/8 patients receiving this dose having drug exposures consistent with anti-tumor activity in pre-clinical models. Fluorodeoxyglucose-positron emission tomography (FDG-PET) scans suggested that GDC-0084 crossed the BBB,

with a uniform distribution throughout the brain. The current phase 2 study will investigate the efficacy and safety of GDC 0084 in patients with newly diagnosed GBM, with unmethylated O₆-methylguanine-methyltransferase (MGMT) promoter status. **METHODS:** This protocol (NCT03522298) has a 2-part design. The phase 1b component consists of an open-label, multicenter dose-escalation study with expansion assess the safety, tolerability, RP2D, PK and clinical activity of GDC 0084 in patients with newly-diagnosed GBM with unmethylated MGMT. The dose-escalation portion of the study will use a standard “3 + 3” design to determine the MTDs for QD, QOD and 3 days on/4 days off schedules. At each of the identified MTDs 10 subjects will be recruited in an expansion cohort. The RP2D will be selected from the three dose schedules being tested. The phase 2 component will evaluate clinical activity of GDC-0084 at the RP2D vs temozolomide as adjuvant therapy following surgical resection and chemoradiation in 224 patients.

ACTR-52. TUMOR TREATING FIELDS (TTFIELDS) IN COMBINATION WITH LOMUSTINE (CCNU) IN THE EF-14 PHASE 3 CLINICAL STUDY – A SAFETY ANALYSIS

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INTRODUCTION: TTFields are low intensity, intermediate frequency, alternating electric fields. The significant overall and progression-free survival benefit, shown in the EF-14 phase 3 study was seen in all patient-subgroups, independent of e.g. MGMT-promotor methylation-status or age. TTFields were not associated with additional systemic toxicity. At recurrence, patients were allowed to continue TTFields with second-line therapies. Here, we analyzed the safety data of TTFields + lomustine (CCNU) to evaluate safety and feasibility of this combination. **METHODS:** Patients in the EF-14 trial received TTFields until second progression, or for 24 months. Change in chemotherapy regimen was allowed after tumor progression. We compared the patients who received lomustine as second-line chemotherapy in combination with TTFields (n=134) to the patients who received lomustine as monotherapy after first progression (n=39). We compared baseline characteristics and the adverse event profile between the groups. **RESULTS:** Baseline characteristics were well balanced except for less female patients in the lomustine only group (7.7% vs. 22.4%). Median age in the TTFields/lomustine group was 55.5 years (29–83) compared to 50.0 years (19–71) for lomustine alone. The addition of TTFields to lomustine therapy was not associated with any significant increase in systemic adverse events compared to lomustine therapy alone (number of patients with 1 SAE 30 % vs. 31 %) and the distribution, severity and overall incidence of adverse events were not statistically different in patients in the two treatment groups. **CONCLUSION:** The data show that the combination of TTFields and lomustine is safe and feasible. This analysis emphasizes again the strong safety profile of TTFields and the high potential of combining TTFields with other modalities. This data is especially important in light of the recently presented, promising data from a small randomized trial that tested the combination of lomustine + TMZ in newly diagnosed (MGMT promotor-methylated only) glioblastoma patients.

ACTR-53. STEAM / EORTC 1608: STUDY OF TG02 IN ELDERLY NEWLY DIAGNOSED OR ADULT RELAPSED PATIENTS WITH ANAPLASTIC ASTROCYTOMA OR GLIOBLASTOMA - A PHASE 1B STUDY

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BACKGROUND: TG02 is an oral multi-cyclin dependent kinase (CDK) inhibitor that is thought to inhibit tumor growth mainly through CDK9-dependent depletion of oncoproteins such as MCL-1 and MYC. MCL-1 and MYC are frequently overexpressed in glioblastoma (up to 80%). Several studies including analyses of patient-derived glioma cell lines have demonstrated profound inhibitory activity (IC₅₀ range = 25–150 nM). Clinical pharmacokinetics from a phase 1b study in multiple myeloma patients demonstrate that TG02 exposures in humans are sufficient for achieving inhibitory concentrations required in the majority of the glioma cell lines tested. Preclinical studies in mice have demonstrated that TG02 is a good candidate for development in gliomas. **METHODS:** Based on these data, the EORTC Brain Tumor Group, in cooperation with Tragara, is currently conducting the phase 1b STEAM trial (EORTC 1608), a three parallel group (A,B,C) open-label, non-randomized, multicenter study. The recommended phase II dose of TG02 in elderly patients with IDH1R132H-non mutant newly diagnosed glioblastoma will be determined in a classical 3 + 3 design dose-escalation and safety study of TG02 in combination with either hypofractionated radiotherapy (RT) or temozolomide (TMZ) in arms A and

B. Patient allocation to treatment in arms A and B will be determined by MGMT promoter methylation status centrally determined according to EANO guidelines for the treatment of elderly patients with glioblastoma. Arm C will explore single agent TG02 activity in anaplastic astrocytoma or glioblastoma at first relapse after initial treatment with TMZ/RTTMZ with a primary endpoint of progression-free survival at 6 months. Secondary objectives include efficacy, quality of life, safety, and correlation with molecular markers. The study is currently open in 2 sites and shall be opened until SNO in 5 additional EORTC sites in Europe. Patient enrolment is planned to start in June 2018. An update will be provided at the SNO conference. NCT03224104

ACTR-54. PATTERN OF RECURRENCE AND IMPACT ON SURVIVAL OF SALVAGE THERAPIES IN LOW-GRADE GLIOMAS FOLLOWING DOSE-DENSE TEMOZOLOMIDE

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INTRODUCTION: Initial chemotherapy with temozolomide (TMZ) may provide some benefit in high-risk low-grade gliomas. No standard treatment is validated at progression. The aim of the study was to investigate which is the optimal salvage therapy after the first progression in terms of PFS and the overall survival (OS). **PATIENTS AND METHODS:** We evaluated 49 patients with grade II oligodendroglioma and oligoastrocytoma according to WHO 2007 included in a phase II AINO trial, who progressed following initial chemotherapy alone with TMZ. Molecular data were available in 48 patients: 29 patients were IDH 1/2 mutated, 23 1p/19q codeleted, and 36 MGMT methylated. Median follow up was 140 months. **RESULTS:** All patients had local tumor progression and patterns on MRI were as follows: a FLAIR lesion in 25 patients (51.0%), a mild patchy enhancing lesion in 14 (28.6%), and a nodular enhancing lesion in the other 10 (20.4%). Twenty-four patients (49%) underwent a second-line chemotherapy, 12 (24.5%) a salvage radiotherapy, 11 (22.4%) a second surgery (10 gross-total resection and 1 subtotal resection), and 2 (4.1%) palliative care. Responses (RANO criteria) following salvage radiotherapy or chemotherapy consisted in PR in 6/38 patients (15.8%), MR in 9/38 (23.7%), SD in 10/38 (26.3%), and PD in 13/38 (34.2%). Median PFS after first salvage therapy was 18 months (IC95% 11–27). Median PFS was 11 months (IC95% 8–74) after radiotherapy, 14 months after chemotherapy and 31 months after surgery (IC95% 18–51, p 0.013). Median OS from the first salvage therapy was 63 months (IC95% 4 - NR). Median OS was 44 months (IC95% 4 NR) after radiotherapy, 38 months (IC95% 22–80) after chemotherapy, and 87 months after surgery (IC95% 11-NR, p 0.09). **CONCLUSIONS:** Reoperation aiming at total or near-total resection seems to offer a probability of a longer PFS and OS compared with the other treatment options.

ACTR-55. TUMOR VOLUME AS A PREDICTOR OF RESPONSE TO ANTI-EGFR ADC ABT-414

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BACKGROUND: Adverse biophysical factors (abnormal vessels and increased interstitial pressure) are increased in larger brain tumors, which negatively impacts penetration of antibody drug conjugates (ADC). The tumour-specific anti-EGFR ADC, depatuximab mafadotin (depatux-m), demonstrated encouraging activity in a Phase 1 glioblastoma study (M12-356 study, NCT01800695). The impact of tumour size on depatux-m efficacy was investigated. **PRECLINICAL STUDY:** Forty mice were engrafted with patient derived xenografts from an M12-356 patient. Eight mice were imaged in a zirconium-labelled depatux-m (3mg/kg) biodistribution study at small (70mm³, n=2) or large (350mm³, n=2) tumour volumes; another 4 mice were imaged with a control ADC. In the controls, non-specific uptake due to blood pool activity was 5% ID/g. In the 89Zr-depatux-m treated groups, mice with larger tumours had significantly less uptake compared smaller tumours (11 vs 21 %ID/gram, p<0.0001). Similarly, tumour inhib-

ition due to depatux-m (3mg/kg every 3 days for 3 weeks, n=16 mice) was significantly less when treatment commenced at larger volumes (27% vs 93%, p<0.001). **CLINICAL STUDY:** To confirm our preclinical data, two reviewers (EL, AS) retrospectively undertook a blinded volumetric analysis of baseline tumor volumes in M12-356. Inter-rater agreement was excellent. Response was compared by chi-square analysis and OS by log-rank. 110 patients with EGFR-amplified, recurrent glioblastoma were treated with depatux-m, alone or in combination with temozolomide. Patients with larger tumours had significant worse response rates (0% in > 25cm³ vs 17% in < 25cm³, p=0.009) and worse overall survival (0.52 vs 0.81 years, p=0.001). The results were similar in patients treated with depatux-m monotherapy for response (n=60, 0% vs 10%, p=0.287) and survival (n=62, 0.50 vs 0.89 years, p=0.001). **CONCLUSIONS:** Increased tumour volumes results in significant reduction in ADC penetration. The impact of this as a modifiable factor, within the broader prognostic impact of increased tumour volume, warrants further investigation with prospective and/or randomized data.

ACTR-57. RELATIVE CEREBRAL BLOOD VOLUME (RCBV) AS RESPONSE PREDICTOR IN THE TREATMENT OF RECURRENT GLIOBLASTOMA (GB) WITH ANTIANGIOGENIC THERAPY

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INTRODUCTION: Glioblastoma (GB) is the most common primary brain tumor in adults having an overall survival (OS) ranging from 12–14 months. GB is highly vascularized, partly due to excessive levels of VEGF. Recurrence is the rule in GB, the second line of treatment is antiangiogenics of which Bevacizumab (BVZ) - a humanized monoclonal antibody to the VEGF-A - is the most commonly used antiangiogenic agent in the setting of recurrent glioblastoma (rGB). rCBV has been evaluated as a predictor of response to this therapy using perfusion-weighted MR techniques. **METHODS:** Patients with rGB according to RANO criteria treated with BVZ with an MRI perfusion with rCBV prior treatment initiation were enrolled. The correlation between rCBV and response was analyzed by correlation coefficient. A cut-off point to identify responders and non-responders to BVZ was determined by ROC curve and OS was evaluated using Kaplan Meier curve and log rank. **RESULTS:** 31 patients with GB were included. We divided the patients in non-responders (n=17) and responders (n=14), there were differences between the groups (age and KPS). The correlation between the RCBV and the Therapeutic response was r=-0.83, p0.0001 (95% CI -0.918 to -0.675). The ROC curve showed 3.7 as the cutoff value with a sensitivity of 100% and specificity of 94% to predict response to BVZ. We divided the groups into non-responders and responders, with OS of 7.7 months vs 14.4 months respectively, p 0.045 (HR 2.3, 95% CI 1.0192 to 5.6192). **CONCLUSIONS:** rCBV is a potential biomarker evaluated by MRI that predicts response to treatment in patients who will be treated with Bevacizumab. Bevacizumab treatment is highly cost-effective, and this biomarker would help identify patients that would benefit. This biomarker as predictive/prognostic has been widely used and is currently the most reliable parameter for measuring the response to antiangiogenic treatment.

ACTR-58. TREATMENT OUTCOMES OF STEREOTACTIC RADIO THERAPY COMBINED WITH BEVACIZUMAB THERAPY IN RECURRENT GLIOBLASTOMA PATIENTS

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BACKGROUND: The prognosis for patients with recurrent glioblastoma is still poor with a median survival between 3 and 6 months. **METHODS:** We performed a retrospective analysis of 11 patients with recurrent glioblastoma treated with stereotactic radiotherapy (25Gy/5fr) followed by 200 mg/m² of Bevacizumab every three weeks. Overall survival was estimated by the Kaplan-Meier method. **RESULTS:** The median age of the group was 59 years. Median overall survival from initial diagnosis was 21 months weeks. Median overall survival from recurrence was 11 months. No adverse events (National Cancer Institute Common Terminology Criteria for Adverse Events grade 3 or 4) occurred in any patient. **CONCLUSION:** Stereotactic radiotherapy combined with BEV treatment was found to be safe and effective in recurrent glioblastoma patients.

ACTR-59. IMPROVING THE INTRA-OPERATIVE DIAGNOSIS OF HIGH-GRADE GLIOMA USING A FLUORESCENCE BIOMARKER - RESULT OF THE GALA-BIDD STUDY

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